(Write your Enrollment Number) Enrollment No._____

MID-TERM EXAMINATION
(B.Tech. CSE-AI/AI &ML)
1st semester
(January, 2023) OFF LINE mode

Subject Code: BAI-103	Subject: Computer Organization and Architecture

Time: 1 ½ Hours Maximum Marks: 30

Q1 Attempt all Parts (2.5*4)

(a) Simplify the following expression using Boolean Algebra:

$AB + A(CD + DC')$

(b) Convert $(372)_{10}$ and $(10110101)_2$ to its equivalent octal number.

(c) A system(A) contains two identical circuits which are operating in parallel. As long as both identical circuits are producing same output, A is producing high output. However, if one of the identical circuits fails, the output of Awill be opposite to the previous output. Considering the identical circuit as a block, design the combination circuit diagram of A which can monitor and detect thata failure has occurred in one of the identical circuits.

(d) Explain the different components of instruction format.

(5,5)

Q2 (Attempt any Two Parts) UNIT-1

(a)Determine by the means of truth table the validity of the following expressions:

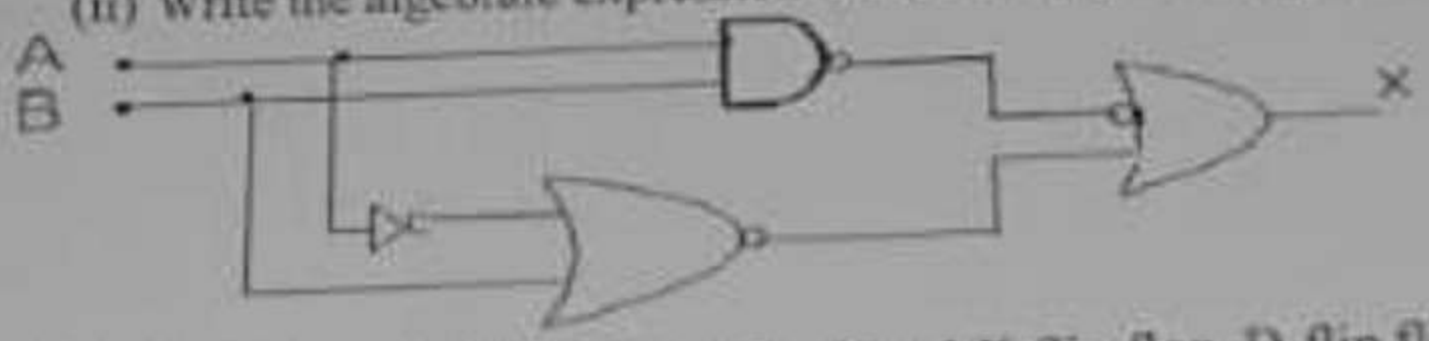
(i) $\bar{A}(A + B) + (B + AA)(A + \bar{B}) = A + B$

(ii) $A + \bar{A}B = A + B$

(b)

(i) Whether NOT logic gate supports associative law or not? Justify.

(ii) Write the algebraic expression for X for the given circuit:



(c) Identify the input states of S-R flip flop, J-K flip-flop, D-flip flop & T-flip flop, which make them act as a buffer/memory flip-flop.

(5,5)

Q3 (Attempt any Two Parts) UNIT-2

(a) Describe the different addressing modes in the instruction code with an example.

(b) Explain the different reference instructions with an example

(c) Consider a hypothetical computer architecture design where 50 different operations are to be defined, and the memory is having 8192 different addresses. Identify the number of bits required for the instruction code in the single address instruction? Also, calculate the number of bits required for **opcode** and **operand memory address** in the indirect mode.